Zhihao Wang

Postdoctoral Researcher, University of Maryland, College Park

GeoAI / GIScience | AI for Science

zhwang0.github.io | Google Scholar | GitHub

Education

University of Maryland, College Park Ph.D., Geographical Sciences 2020–2026

Advisor: Yiqun Xie.

Dissertation: *Deep Learning Emulation for Scalable Ecosystem Forecasting: Methods, Global Benchmarks, and Tipping-Point Discovery.*

The Ohio State University M.A., Geography 2018–2020

University of Waterloo, Canada B.E.S., Honours Geomatics 2016–2018

Minor in Computer Science.

Wuhan University, China B.E., Remote Sensing Science and Technology 2014–2018

Academic & Research Appointments

Postdoctoral Researcher Jun 2026–Present

Department of Geographical Sciences, University of Maryland, College Park.

Graduate Research Assistant 2020–2026

Department of Geographical Sciences, University of Maryland, College Park.

Outstanding Graduate Assistant Award, 2026 (approximately 80 recipients among 4,000+ graduate assistants).

Graduate Research Assistant May 2021–Aug 2022

National Oceanic and Atmospheric Administration (NOAA).

Developed albedo climatology for the NOAA-20 VIIRS Global Albedo product using 3.5+ TB of satellite observations; contributed memory-efficient processing and operational algorithm testing.

Research Interests

- **Scientific machine learning and world models:** physics-guided learning, neural emulation, spatiotemporal forecasting, and what-if simulation.
- **Geospatial AI and foundation models:** computer vision, model evaluation, and geographic generalization.
- **Reliable geospatial learning:** uncertainty quantification, geographic fairness, benchmark design, and robustness under distribution shift.

Publications

AI/Computer Science Conference Proceedings (Peer-Reviewed)

[C1] **Wang, Z.**, Wang, R., Li, R., Ma, L., Hurtt, G., Jia, X., Mai, G., Wang, S., & Xie, Y. (2026). Earth System World Model for What-If Simulations: A Case Study for Terrestrial Ecosystems. *ACM SIGSPATIAL*.

[C2] **Wang, Z.**, Ma, L., Hurtt, G., Jia, X., Li, Y., Li, R., Li, Z., Xu, S., & Xie, Y. (2025). CarbonGlobe: A Global-Scale, Multi-Decade Dataset and Benchmark for Carbon Forecasting in Forest Ecosystems. *NeurIPS*.

[C3] **Wang, Z.**, Li, C., Wang, R., Ma, L., Hurtt, G., Jia, X., Mai, G., Li, Z., & Xie, Y. (2025). TreeFinder: A US-Scale Benchmark Dataset for Individual Tree Mortality Monitoring Using High-Resolution Aerial Imagery. *NeurIPS*.

[C4] **Wang, Z.**, Xie, Y., Li, Z., Jia, X., Jiang, Z., Jia, A., & Xu, S. (2024). SimFair: Physics-Guided Fairness-Aware Learning with Simulation Models. *AAAI*, 38(20), 22420–22428.

[C5] **Wang, Z.**, Xie, Y., Jia, X., Ma, L., & Hurtt, G. (2023). High-Fidelity Deep Approximation of Ecosystem Simulation over Long-Term at Large Scale. *ACM SIGSPATIAL*, pp. 1–10. **Oral Presentation**.

[C6] Li, R., **Wang, Z.**, Xu, S., Li, Z., Wang, R., Ma, L., Hurtt, G. C., Jia, X., Mai, G., & Xie, Y. (2026). Fast-EcoDiffusion: Scalable Uncertainty-Aware Emulation of Ecosystem Processes via Latent Diffusion Models. *ACM SIGSPATIAL*.

[C7] Xu, S., **Wang, Z.**, Li, R., Wang, R., Ma, L., Hurtt, G. C., Jia, X., & Xie, Y. (2026). Knowledge-Guided Learning for Global Carbon Flux Prediction: Integrating High-Level Remote Sensing with Bottom-Up Physical Modeling. *ACM SIGKDD*, pp. 12561–12572.

[C8] Li, R., **Wang, Z.**, Jia, X., Mai, G., Ma, L., Hurtt, G. C., Shen, Q., Li, Z., & Xie, Y. (2026). EcoDiffusion: Uncertainty-Aware Emulation of Ecosystem Processes with Conditional Diffusion for Long Sequences with Single-Step Initialization. *AAAI*, 40(45), 38880–38888.

[C9] Li, R., Xie, Y., Jia, X., Mai, G., Hou, S., **Wang, Z.**, & Li, Z. (2025). Scenario-Based Evaluation of Probabilistic Time Series Forecasting for Solar Energy. *ACM SIGSPATIAL*, Short Paper (Experiment), pp. 631–634.

[C10] Li, R., Xie, Y., Jia, X., Wang, D., Li, Y., Zhang, Y., **Wang, Z.**, & Li, Z. (2024). SolarCube: An Integrative Benchmark Dataset Harnessing Satellite and In-situ Observations for Large-scale Solar Energy Forecasting. *NeurIPS*.

Journal Articles (Peer-Reviewed)

[J1] Li, Z., Xie, Y., Skakun, S., Jia, X., Mai, G., Lu, W., Tong, M., & **Wang, Z.** (2026). Annotation-free cloud masking for PlanetScope images in the Arctic via cross-platform ability transfer using deep learning and foundation models. *Remote Sensing of Environment*, 334, 115138.

[J2] Li, R., Wang, D., **Wang, Z.**, Liang, S., Li, Z., Xie, Y., & He, J. (2025). Transformer approach to nowcasting solar energy using geostationary satellite data. *Applied Energy*, 377, 124387.

[J3] Li, Z., Xie, Y., Jia, X., Mai, G., **Wang, Z.**, & Chen, W. (2025). Machine-learning-enabled spatial pattern mining: evaluating the impact of imperfect inputs. *International Journal of Geographical Information Science*, 39(10), 2362–2400.

[J4] Xie, Y., Nhu, A. N., Song, X. P., Jia, X., Skakun, S., Li, H., & **Wang, Z.** (2025). Accounting for spatial variability with geo-aware random forest: A case study for US major crop mapping. *Remote Sensing of Environment*, 319, 114585.

[J5] Chen, D., Fu, C., Jenkins, L., He, J., **Wang, Z.**, Jandt, R., Frost, G., Bredder, A., Berner, L., & Loboda, T. (2024). Regional fire-greening positive feedback loops in Alaskan Arctic tundra. *Nature Plants*, 10, 1886–1891.

[J6] Xu, S., Wang, D., Liang, S., Jia, A., Li, R., **Wang, Z.**, & Liu, Y. (2024). A novel approach to estimate land surface temperature from Landsat top-of-atmosphere reflective and emissive data using transfer-learning neural network. *Science of the Total Environment*, 955, 176783.

[J7] Peng, J., Yu, P., Yu, Y., Jia, A., Wang, D., Wang, H., & **Wang, Z.** (2023). An evaluation of the NOAA global daily gap-filled VIIRS surface albedo. *Remote Sensing of Environment*, 298, 113822.

[J8] Jia, A., Liang, S., Wang, D., Ma, L., **Wang, Z.**, & Xu, S. (2023). Global hourly, 5 km, all-sky land surface temperature data from 2011 to 2021 based on integrating geostationary and polar-orbiting satellite data. *Earth System Science Data*, 15(2), 869–895.

[J9] Liang, X., Liu, D., **Wang, Z.**, & Wang, J. (2022). Characterizing the dynamics of wildland-urban interface and the potential impacts on fire activity in Alaska from 2000 to 2010. *Landscape and Urban Planning*, 228, 104553.

[J10] Li, R., Wang, D., Liang, S., Jia, A., & **Wang, Z.** (2022). Estimating global downward shortwave radiation from VIIRS data using a transfer-learning neural network. *Remote Sensing of Environment*, 274, 112999.

[J11] Li, Y., Liu, Y., Bohrer, G., Cai, Y., Wilson, A., Hu, T., **Wang, Z.**, & Zhao, K. (2022). Impacts of forest loss on local climate across the conterminous United States: Evidence from satellite time-series observations. *Science of the Total Environment*, 802, 149651.

Workshop, GISCup & Proceedings Papers

* Equal contribution.

[W1] Li, C., **Wang, Z.**, & Xie, Y. (2025). Multimodal Foundation Models as Router Models for High-Resolution Aerial Image Segmentation. *IEEE International Conference on Data Mining Workshops (ICDMW)*.

[W2] **Wang, Z.**, Li, Z., Xie, Y., Souza Jr, C., Souza Filho, J. S., & Pinheiro, S. (2024). AI-based Validation of Deforestation Using High-Resolution Satellite Imagery in the Brazilian Amazon. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLVIII-3-2024, 583–588.

[W3] Chen, W.*, **Wang, Z.***, Li, Z.*, Xie, Y., Jia, X., & Li, A. (2022). Deep semantic segmentation for building detection using knowledge-informed features from LiDAR point clouds. *ACM SIGSPATIAL*, GISCup paper, pp. 1–4. **Third Place, GISCup**.

Preprints & Manuscripts

[M1] **Wang, Z.**, et al. (2026). Re-Imagining Foundation Models for Earth Monitoring: From Larger Models and Benchmark Gains to Operational Value. *Manuscript in preparation.*

[M2] Mai, G., Rao, J., Cao, Q., Liu, Z., **Wang, Z.**, et al. (2026). Towards Generative Interactive Geospatial World Models for Proactive and Scenario-Driven Geospatial Decision-Making. *Manuscript under review.*

[M3] Xie, Y., **Wang, Z.**, Chen, W., Li, Z., Jia, X., Li, Y., Wang, R., Chai, K., Li, R., & Skakun, S. (2024). When are Foundation Models Effective? Understanding the Suitability for Pixel-Level Classification Using Multispectral Imagery. *arXiv:2404.11797*.

Honors & Awards

First Place (tie), UCGIS Student Lightning Talk Award	2026
Outstanding Graduate Assistant Award, University of Maryland	2026
NeurIPS Scholar Award (travel)	2025
AAAI Student Scholarship (travel)	2024
Second Place, Excellence in Graduate Research Award	2024
Department of Geographical Sciences, University of Maryland	
ACM SIGSPATIAL GISCup Travel Award	2023
Third Place, ACM SIGSPATIAL GISCup Competition	2022
Dean's Fellowship, University of Maryland	2020
Dean's Honor List and Entrance Scholarship, University of Waterloo	2016, 2017, 2018
Wuhan University Scholarship and 5050 Scholarship	2016

Teaching Experience

Instructor — University of Maryland

GEOG 276: Principles of Python Programming and Geocomputing	2025

Guest Lecturer — University of Maryland

GEOG 788T: Deep Learning for Spatial and Spatio-Temporal Data	2025
GEOG 398E: Introduction to Spatial Artificial Intelligence	2025

Teaching Assistant

GEOG 140: Natural Disaster, University of Maryland 2020

GEOG 5210: Fundamentals of GIS, The Ohio State University 2020

GEOG 5201: Geo-visualization, The Ohio State University 2019

GEOG 5225: Geographic Applications of Remote Sensing, The Ohio State University 2019

GEOG 5226: Spatial Simulation and Modeling in GIS, The Ohio State University 2018

Selected Presentations

- **Wang, Z.** (2026). TreeFinder: A US-Scale GeoAI Benchmark for Individual Tree Mortality Monitoring. University Consortium for Geographic Information Science (UCGIS) Symposium. **First Place (tie), Student Lightning Talk Award.**
- **Wang, Z.** (2026). Scalable AI for US-Scale Individual Tree Mortality Monitoring. Workshop on AI & Analytics for Social Good, College Park. **Oral; one of five selected Ph.D. presentations.**
- **Wang, Z.** (2026). CarbonGlobe: A Global-Scale, Multi-Decade Dataset and Benchmark for Carbon Forecasting in Forest Ecosystems. Inaugural AIM Research & Learning Symposium, College Park.
- **Wang, Z.**, Xie, Y., Jia, X., Ma, L., Ott, L., & Hurtt, G. (2025). Deep-ED: A Deep Learning Emulator for Uncovering Ecosystem Tipping Points under Intermediate Climate Scenarios. American Geophysical Union (AGU) Annual Meeting, New Orleans. **Oral.**
- **Wang, Z.**, Xie, Y., Jia, X., Ma, L., Ott, L., & Hurtt, G. (2024). Deep-ED: An Advanced Deep Learning Approach for Ecosystem Dynamics and Global Carbon Forecasting. American Geophysical Union (AGU) Annual Meeting, Washington, D.C.
- **Wang, Z.**, & Liu, D. (2019). A Spatial-Temporal Classification Method Based on Convolutional Neural Network and Markov Random Field. American Geophysical Union (AGU) Annual Meeting, San Francisco.
- **Wang, Z.**, & Liu, D. (2019). A Method to Optimize Temporal Sequence and Spatial Context in Multi-Temporal Land Cover Maps. Association of American Geographers (AAG) Annual Meeting, Washington, D.C.

Service & Research Leadership

Organizing Committees

- IEEE BigData Cup: AI Emulation Challenge. 2026
- ACM SIGSPATIAL International Workshop on Spatial Big Data and AI for Industrial Applications (GeoIndustry), 4th and 5th editions. 2025, 2026

Program Committees

- AAAI 2026, 2027
- ACM SIGKDD 2025 (Cycle 2), 2026 (Cycle 1)
- NeurIPS 2026
- ACM SIGSPATIAL 2025, 2026
- ACM SIGSPATIAL UMFM 2026
- IEEE BigData 2025

Journal Reviewing

Remote Sensing of Environment; IEEE Transactions on Geoscience and Remote Sensing; International Journal of Applied Earth Observation and Geoinformation; GIScience & Remote Sensing; Neurocomputing; IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing; International Journal of Digital Earth; Scientific Reports; ACM Transactions on Knowledge Discovery from Data; Humanities & Social Sciences Communications; Global Transitions; European Journal of Remote Sensing; Environmental Monitoring and Assessment; Natural Hazards; GeoInformatica.

Additional Conference Reviews under Advisor Supervision

AAAI (2023, 2024, 2025); IJCAI (2023, 2024, 2025); KDD (2025); ACM SIGSPATIAL (2023, 2024); ICDM (2025); IEEE BigData (2024); SDM (2023).

Student Mentoring

High school research mentoring.

- Cooper Li, Montgomery Blair High School 2024, 2025
- Krish Pruthi, Poolesville High School 2025
- Peter Jin, James M. Bennett High School 2025